Motor and Fire Claims Performance Review

Why did motor loss ratios jump after the 2023 devaluation, and how much price do we need for 2027?

A portfolio project in business performance analysis for a Malawi short-term insurer
Period covered: January 2023 to September 2026

Summary

The Motor insurance made less money after the kwacha was devalued in November 2023. We wanted to know where the problem is, what caused it, whether repricing has worked, and how much more price is needed in 2027.

What we found

- **Motor loss ratios jumped and have now mostly recovered.** The combined motor loss ratio was 57.7% in 2023, 82.8% in 2024 and 74.5% in 2025. For January to September 2026 it is 63.2%, close to the 60% target.
- **Bigger claims caused the problem, not more claims.** Claims per 100 policies stayed at about 12 to 13 every year. The average motor private claim grew from K1,398,312 in 2023 to K3,292,672 in 2026, which is 135% more. Average premium grew 114% over the same time.
- **Premium fell behind inflation in 2024 and caught up in 2025.** After allowing for inflation, premium growth was -5.4% in 2024, then +3.9% in 2025 and +4.3% in 2026 so far.
- **One segment still needs attention.** Commercial motor in Lilongwe ran at 77.0% over the last 12 months, the only branch and product pair above 70%.
- **Customers are leaving as prices rise.** The yearly lapse rate for motor private went from 23.2% in 2023 to 29.6% in 2026.
- **Claims are slow in the smaller branches.** The median time to settle a claim is 35 days in Zomba and 31 days in Mzuzu, against 22 in Blantyre.

What we recommend

- Raise motor rates in 2027 by about 15% (private) and 25% (commercial) on top of the automatic rise from sum-insured revaluation. Phase it in over two renewal rounds and start with Lilongwe commercial.
- Work on the cost of each claim: agreed garage rates, a parts price list reviewed every quarter, and a repair network check.

- Protect customers before each price rise: early contact, payment by instalments on mobile money, and a monthly lapse report.
- Cut settlement time in Zomba and Mzuzu with more claims staff or digital claim intake, and review the claims flagged as unusual.
- Do not cut fire prices yet. The 12-month result rests on only a few claims.

2. About the data

Please read: the policy and claims data are simulated

No real company data is used in this project. The data was simulated by real, public facts: the inflation path, the November 2023 devaluation, the exchange rate and the policy rate (listed in the table below). The claim sizes, premiums and branch results are assumptions.

Because the data is simulated, the findings show how to do the analysis. They are not statements about any real insurer.

2.1 The three data files

File	Rows	What it holds
monthly_performance.csv	720	One row for each month, branch and product (45 months x 4 branches x 4 products). Policies in force, new and lapsed policies, earned premium, claim count, claims incurred, commission and management expenses.
claims.csv	9,938	One row for each claim: loss date, branch, product, cause, how it was reported, amount, days to report, days to settle, status (settled or open) and a flag for unusual claims.
macro_indicators.csv	45	Monthly inflation, official kwacha per US dollar and the policy rate. Some inflation months are estimates, marked in the file.

2.2 Public facts used to shape the data

Fact	Value	Source and date
Annual inflation, September 2026	19.4% (August: 20.0%)	National Statistical Office, as reported by The Rio Times, 19 Oct 2026
Annual inflation, January to July 2026	24.9%, 24.1%, 23.8%, 24.3%, 23.4%, 21.1%, 20.8%	NSO figures as reported in the press (Trading Economics, AllAfrica, Malawi24)

Annual inflation, September 2025	28.7%	NSO Consumer Price Index release, September 2025
Kwacha devaluation, November 2023	From about K1,180 to K1,700 per US dollar	Reuters, reported by Daba Finance, 9 Nov 2023
Official exchange rate, October 2026	About K1,734 per US dollar	Reserve Bank of Malawi indicative rate, 13 Oct 2026, reported by The Rio Times
Policy rate	Cut from 26% to 24% in March 2026, held in August 2026	Reserve Bank of Malawi MPC decisions, reported by MBC and CDH Malawi
Industry gross written premium, 2023	K174.8 billion	GlobalData. Used only to check that the simulated book is a believable size.

3. Findings

3.1 The portfolio at a glance

Year	Earned premium (K bn)	Claims (K bn)	Loss ratio	Expense ratio	Combined ratio	Policies in force at end
2023	11.7	6.3	53.9%	29.3%	83.3%	21,845
2024	14.6	9.7	66.5%	29.8%	96.2%	21,641
2025	19.8	12.5	63.0%	30.0%	93.1%	20,939
2026 (Jan-Sep)	18.7	9.7	52.0%	30.2%	82.2%	20,545

The 2026 column covers nine months only, so its premium and claim totals are smaller. The ratios can still be compared with full years. The portfolio-wide combined ratio rose from 83.3% in 2023 to 96.2% in 2024, close to break-even on underwriting. Fire and personal accident held the total down.

3.2 Premium growth against inflation

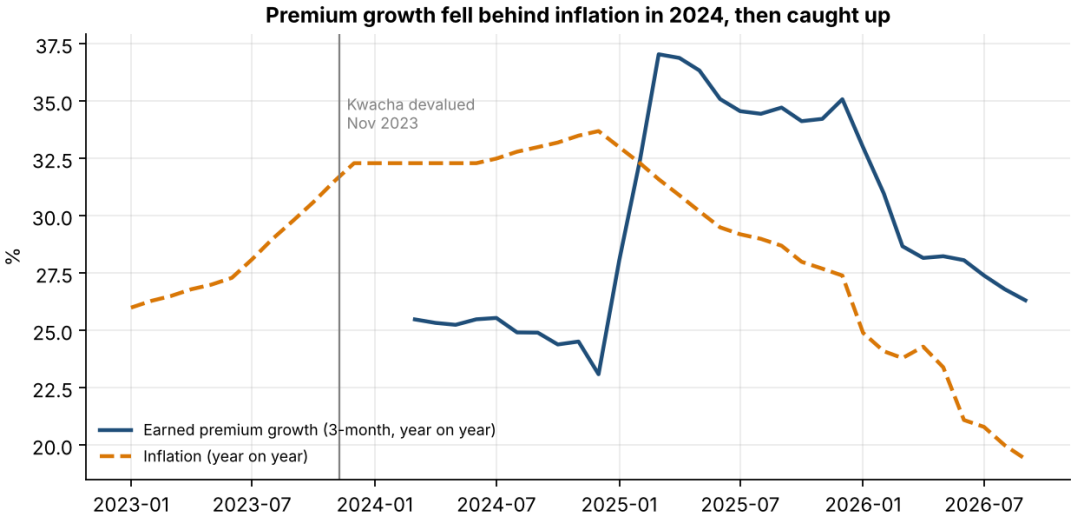


Figure 1. Year-on-year growth in earned premium (3-month basis) against annual inflation.

After the November 2023 devaluation, inflation stayed above 32% through most of 2024. Premium was growing at about 25% a year at that point, so premiums lost ground. The rate increases of January 2025 and January 2026 lifted premium growth above inflation. For January to September, real premium growth was -5.4% in 2024, +3.9% in 2025 and +4.3% in 2026.

3.3 Loss ratios by product

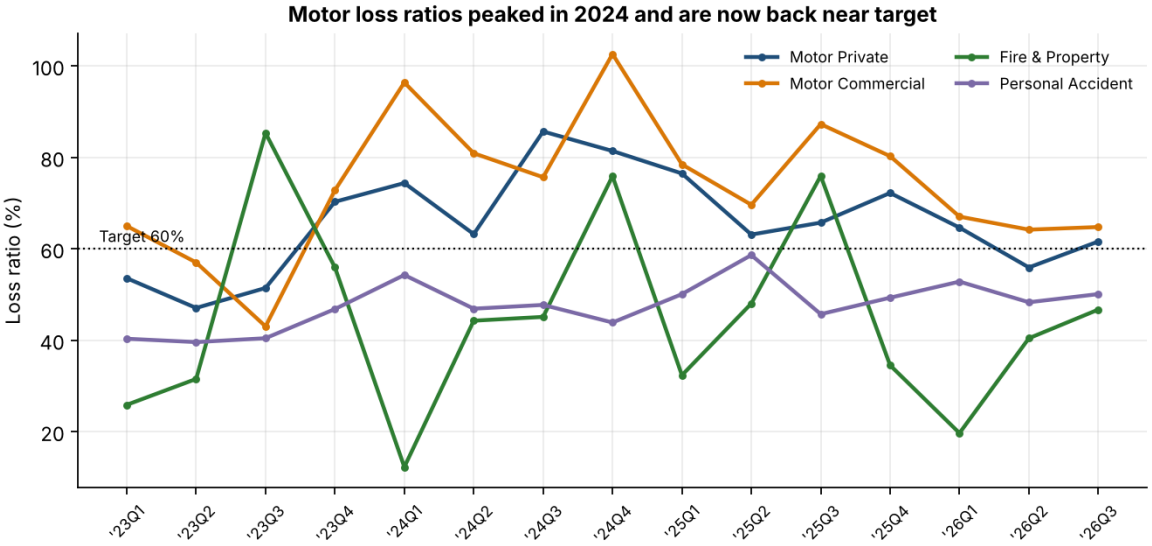


Figure 2. Quarterly loss ratio by product. The dotted line is the 60% target.

Product	2023	2024	2025	2026 (Jan-Sep)
Motor Private	55.8%	76.4%	69.4%	60.7%
Motor Commercial	59.5%	88.9%	79.0%	65.3%
Fire & Property	50.6%	45.4%	47.9%	35.9%

Personal Accident	41.8%	48.1%	50.9%	50.4%

Motor commercial was hit hardest, reaching 88.9% in 2024. Quarterly results for fire swing a lot because a single large claim can move a quarter. Personal accident has stayed stable between 42% and 51%.

3.4 What drove the loss ratio: claim size

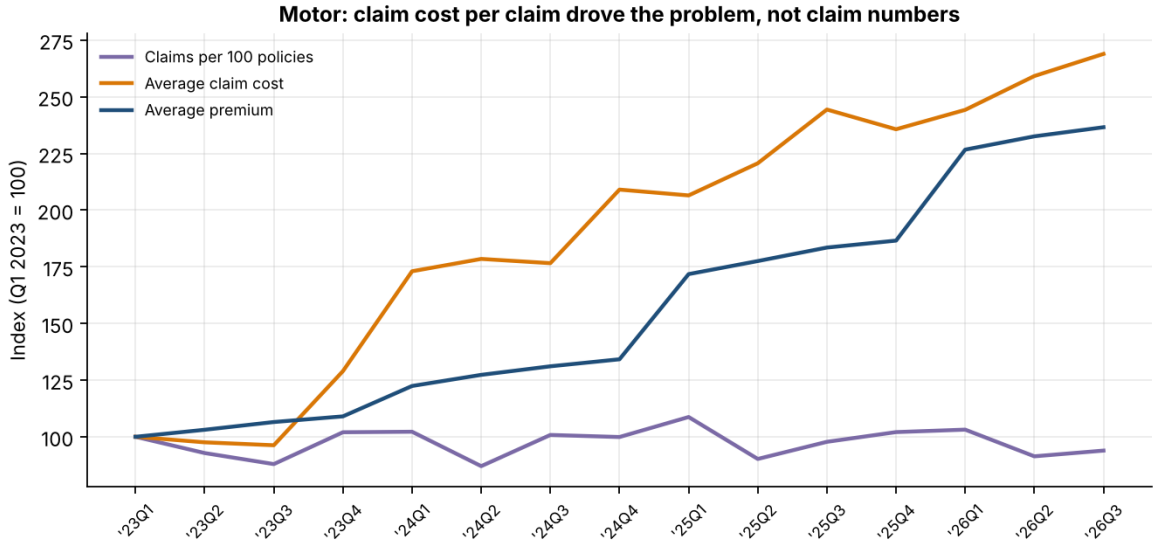


Figure 3. Motor: claims per 100 policies, average claim and average premium, indexed to the first quarter of 2023 (= 100).

Period	**Loss ratio start**	**Loss ratio end**	**More or fewer claims**	**Average claim size**	**Average premium**	**Total change**
2023 to 2024	57.7%	82.8%	+1.9	+55.0	-20.7	+36.2
2024 to 2025	82.8%	74.5%	+2.3	+20.5	-33.4	-10.6
2025 to 2026 (Jan-Sep)	73.8%	63.2%	-2.8	+14.1	-26.7	-15.5

From 2023 to 2024, claim size added about 55 points to the motor loss ratio, while claim numbers added only 2 point. Higher premiums took off about 21 points but did not offset it. In 2024 the average motor private claim was 69% higher than in 2023, and the average motor commercial claim was 73% higher. This fits what the devaluation does: spare parts, tyres and labour linked to imports cost more in kwacha. In 2025 and 2026 premium rises were larger than claim size rises, so the loss ratio came down.

3.5 Where to act: branch and product

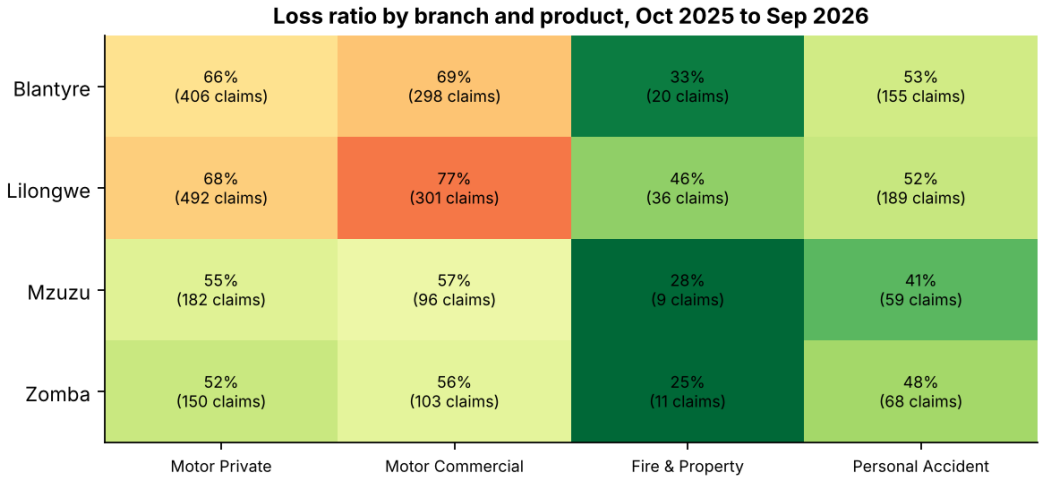


Figure 4. Loss ratio by branch and product for October 2025 to September 2026, with the number of claims.

Lilongwe commercial motor stands out at 77.0% on 301 claims. Lilongwe also has the highest loss ratio of any branch overall (61.6%), while Zomba (42.0%) and Mzuzu (44.1%) are well under target. Fire claims shows very low loss ratios in some branches, but the whole fire book had only 76.0 claims in 12 months, so one large loss would change the picture.

3.6 Customers are leaving

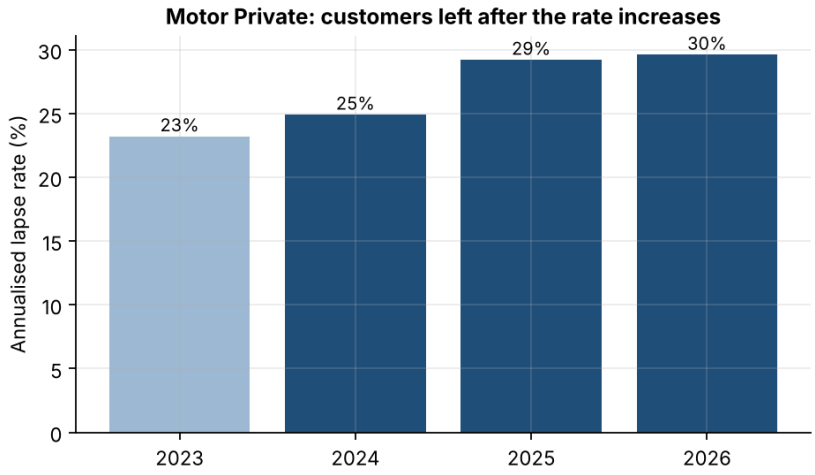


Figure 5. Motor private: yearly lapse rate for January to September of each year.

Lapse for motor private rose from 23.2% to 29.6%. Lapse also rose for motor commercial (19.2% to 21.7%) and personal accident (26.7% to 28.4%). In this data each January rate increase was followed by more lapses in the next six months. The higher the price rise, the bigger the jump. The next price round has to be planned with retention in mind.

3.7 Claims handling



Figure 6. Days from report to settlement, settled claims only.

Branch	Settled claims	Median days	Average days	Share taking over 30 days from the loss report
Blantyre	3,338	22	25.2	35.3%
Lilongwe	3,969	23	27.2	41.1%
Mzuzu	1,227	31	35.9	59.9%
Zomba	1,196	35	40.3	66.8%

Zomba and Mzuzu are the slowest. Two out of three Zomba claims take more than 30 days from report to settlement (66.8%). At the end of September 2026, 208 claims were still open, worth K1.23 bn at the estimated amount. The claims file also flags 2.6% to 2.8% of motor and fire claims as unusual. A flag is a reason to look, not proof of fraud. Claims with a flag total K0.91 bn over the 45 months.

3.8 Statistical checks

Question	Method	Result
Did the motor loss ratio change after the devaluation?	Welch t-test on monthly motor loss ratio: 10 months before (Jan-Oct 2023) against 12 months after (Dec 2023 to Nov 2024)	Average rose from 52.8% to 82.6%. $t = 7.66$, $p < 0.001$. The rise is not chance.
Do claim costs follow inflation?	Regression of year-on-year growth in the average motor claim on inflation and a devaluation-period marker (33 months)	R-squared 0.8. The devaluation period adds about 60 points to claim cost growth ($p < 0.001$).

		Inflation by itself is not significant here (p = 0.26).
Does settlement time differ by branch?	One-way ANOVA on the log of settlement days	F = 292, p < 0.001, using 9,730 settled claims. Branches differ.

3.9 Looking ahead: the price we need for 2027

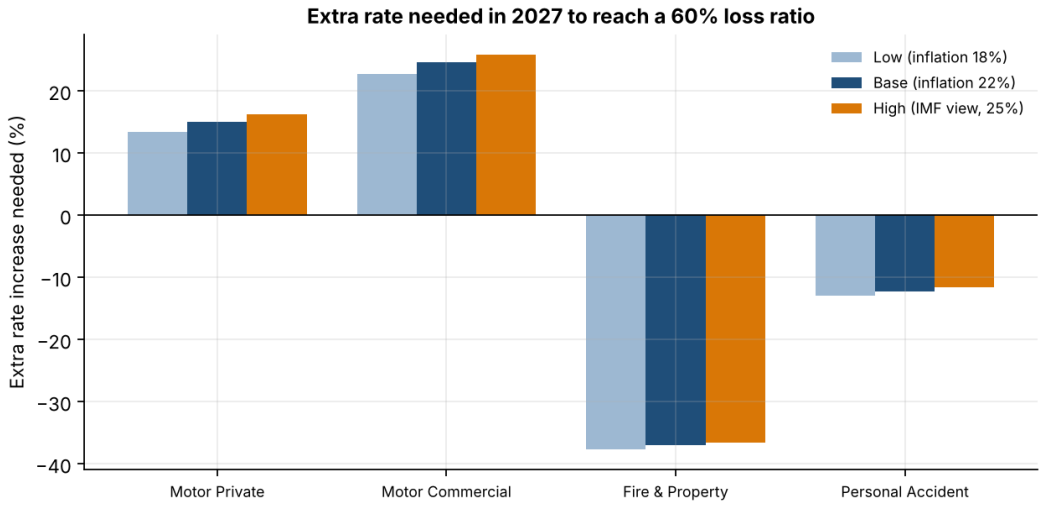


Figure 7. Extra rate increase needed to reach a 60% loss ratio, beyond the automatic premium rise from sum-insured revaluation.

Product	**Loss ratio, last 12 months**	**Inflation 18%**	**Inflation 22% (base)**	**Inflation 25%**
Motor Private	63.2%	+13.4%	+15.1%	+16.3%
Motor Commercial	68.5%	+22.8%	+24.6%	+25.9%
Fire & Property	35.6%	-37.7%	-37.0%	-36.6%
Personal Accident	50.1%	-12.9%	-12.2%	-11.6%

The 22% base case is close to the Reserve Bank of Malawi forecast for 2026 average inflation. The 25% case matches the IMF projection for 2027. Motor private needs about 15% and motor commercial about 25% more than the automatic rise. Fire and personal accident already earn more than the target needs. For fire, the 12-month result rests on very few claims, so hold the rate. For personal accident, the room could be used to keep customers, since its lapse rate is high.

4. Recommendations

Action	**Evidence**	**When**	**How we will know it worked**

Raise motor rates in two steps. Private about 15%, commercial about 25% above the automatic rise. Start with Lilongwe commercial.	Table in 4.9. Lilongwe commercial at 77%.	January and July 2027 renewals	Motor loss ratio at or under 60% on a rolling 12 months
Control the cost of each claim: agreed garage rates, quarterly parts price list, repair network review.	Claim size drove the loss ratio (4.4).	First quarter of 2027	Average motor claim grows no faster than inflation
Protect retention: call customers facing the biggest rises, offer mobile money instalments, report lapse monthly.	Lapse rose after each rate round (4.6).	Before the January 2027 renewals	Motor private lapse back under 25%
Speed up claims in Zomba and Mzuzu: extra staff or digital claim intake (WhatsApp already takes about 10% of reports).	Median 35 and 31 days against 22 (4.7).	First half of 2027	Share of claims over 30 days under 40% in both branches
Set a referral rule for flagged claims and review them before payment above an agreed amount.	Flagged claims total about K0.9 bn over 45 months (4.7).	First quarter of 2027	Share of flagged claims reviewed. Savings recorded.
Hold fire and personal accident rates. Revisit fire when there are more claims to rely on.	Few fire claims (4.5). High lapse in accident (4.6).	Review at mid-2027	Fire loss ratio on a 24-month view
Turn this analysis into a monthly KPI pack for management.	Repeatable method (section 3).	From January 2027	Pack issued by working day 10 each month

5. Limits of this work

- The data is simulated. The method is real; the numbers are not from any insurer. The link between price rises and lapses, and between the devaluation and claim cost, was built into the simulation, so the tests confirm the method works rather than discover a new fact.

- Claims incurred are shown at full estimated amount, including claims that are still open. Late-developing claims for recent months could change the latest loss ratios.
- The expense ratio is an assumption (about 30%). The analysis ignores reinsurance, so loss ratios are gross.
- Some branch and product pairs have few claims. Fire is the clearest case. Results for them are noisy.
- The regression uses 33 months and cannot separate inflation from the devaluation step. The pass-through factors in the 2027 view are assumptions and should be checked against the insurer's own data.

Appendix A. Key measures by product and year

Product	Year	Loss ratio	Combined ratio	Claims per 100 policy-years	Average claim (K m)	Average premium (K thousand per year)
Motor Private	2023	55.8%	84.5%	17.0	1.40	427
	2024	76.4%	105.3%	16.9	2.36	523
	2025	69.4%	98.6%	17.4	2.86	720
	2026	60.7%	90.3%	16.9	3.29	917
Motor Commercial	2023	59.5%	88.1%	26.9	2.64	1,197
	2024	88.9%	117.8%	28.3	4.57	1,456
	2025	79.0%	108.3%	28.3	5.63	2,018
	2026	65.3%	94.6%	26.7	6.28	2,569
Fire & Property	2023	50.6%	81.6%	3.3	30.28	1,973
	2024	45.4%	76.9%	3.1	34.35	2,334
	2025	47.9%	79.5%	2.8	50.53	2,958
	2026	35.9%	67.8%	2.8	44.98	3,506
Personal Accident	2023	41.8%	67.9%	5.5	0.69	92
	2024	48.1%	74.5%	5.8	0.87	105
	2025	50.9%	77.6%	6.4	1.00	125
	2026	50.4%	77.2%	5.9	1.20	142

2026 covers January to September.

Appendix B. Data dictionary

Column	Meaning
month, branch, product	Month (YYYY-MM), one of four branches, one of four products
policies_in_force	Policies in force at the end of the month
new_policies, lapsed_policies	Policies sold and policies not renewed in the month
earned_premium_mwk	Premium earned in the month, in kwacha

claim_count, claims_incurred_mwk	Number and total value of claims with a loss date in the month
commission_mwk, management_expenses_mwk	Commission paid and management expenses allocated to the month
claim_id, loss_date, loss_month	Claim reference and date of loss
cause, reporting_channel	Cause of loss and how the claim was reported (branch, agent or broker, call centre, WhatsApp)
claim_amount_mwk	Claim amount, in kwacha, capped at a net retention limit
report_lag_days, settlement_days	Days from loss to report, and days from report to settlement
status, fraud_flag	Settled or open on 30 September 2026. 1 if the claim was flagged as unusual.
inflation_yoy_pct, usd_mwk_official, policy_rate_pct	Macro series. inflation_source says whether each inflation month is a published figure or an estimate.